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## MEMORY DEVICE WITH REDUNDANCY FOR PAGE-BASED REPAIR

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### Abstract

Apparatus and methods for page-based soft post package repair are disclosed. Based on data stored in a storage element, an address may be decoded to a prime row, a row-based redundant row, or a page-based redundant row. A match logic circuit may determine whether the address corresponds to a defective prime row and generate a match signal. A decoder can select a redundant row to be accessed instead of a prime row in response to the match signal indicating that the address data corresponding to the address to be accessed matches defective address data stored in a volatile memory. A page-based redundant row allows for page-by-page substitution for defective memory, allowing functional portions of memory to continue to be used.

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## **Background/Summary**

[0001] This application is a continuation of U.S. application Ser. No. 18/761,619, filed Jul. 2, 2024, which is a continuation of U.S. patent application Ser. No. 17/823,740, filed Aug. 31, 2022, now issued as U.S. Pat. No. 12,062,407, all of which are incorporated herein by reference in their entirety. [0002] Embodiments of the disclosure relate generally to memory devices and, more specifically, to memory devices with configurations of redundant rows facilitating improved page-based repair in the memory device.

### **BACKGROUND**

[0003] Memory cells of memory devices such as dynamic random access memories (DRAMs), static RAMs (SRAMs), flash memories, or the like can experience defects leading to errors and/or failures. In some cases, memory cells can be identified as defective (hereinafter “defective memory cells”) during the manufacturing process, including after the memory device (e.g., a memory chip) has been packaged, such as in cases where the memory cells were not defective before the packaging process. Examples of packaging include, but are not limited to, encapsulation by epoxy, ceramic packages, metal/glass packages, and the like. After a memory device has been packaged, the memory device can be tested to identify defective memory cells. Addresses mapped (e.g., assigned) to defective memory cells can be remapped (e.g., reassigned) to functional memory cells (e.g., memory cells that have not been identified as defective) so that the memory device can have the designed capacity, or at least improved capacity over the defective device.

[0004] The remapping can be performed using hardware modification. For example, by using antifuses, an address originally mapped to a first row can be remapped to a different row. An antifuse has a high resistance in its initial state. An antifuse can permanently create an electrically conductive path when a relatively high voltage is applied across the antifuse. An antifuse can have a structure similar to that of a capacitor, i.e., two conductive electrical terminals are separated by a dielectric layer. To create an electrically conductive path, a relatively high voltage is applied across the terminals, breaking down the interposed dielectric layer and forming a conductive link between the antifuse terminals. Creating a conductive path through an antifuse is referred to as “blowing an antifuse.”

[0005] Certain protocols exist for performing post package repair. Post package repair can involve blowing antifuses. Blowing an antifuse can be performed in an antifuse programming time, which can be on the order of 200 milliseconds (ms). In some applications, such a delay can undesirably impact performance of a memory.

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## **Description**

### **BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS**

[0006] The disclosure will be understood more fully from the detailed description given below and from the accompanying drawings of various embodiments of the disclosure. The drawings, however, should not be taken to limit the disclosure to the specific embodiments, but are for explanation and understanding only.

[0007] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0008] FIG. 1 is a block diagram illustrating a memory bank of a memory device configured to

remap memory addresses using row-based repairs and page-based repairs, according to an embodiment.

[0009] FIG. 2 is a flow chart showing operations of a method performed by a circuit in remapping memory address, according to an embodiment.

[0010] FIG. 3 is a block diagram illustrating page-based repairs, according to an embodiment.

[0011] FIG. 4 is a block diagram illustrating an example row decoder, according to an embodiment.

[0012] FIG. 5 is a block diagram illustrating an example row decoder, according to an embodiment.

[0013] FIG. 6 is a block diagram illustrating a soft post-package repair (SPPR) circuit, according to an embodiment.

[0014] FIG. 7 is a block diagram illustrating a redundant row decode circuit, according to an embodiment.

[0015] FIG. 8 is a flow chart showing operations of a method performed by a circuit in remapping a memory address, according to an embodiment.

[0016] FIG. 9 is a flow chart showing operations of a method performed by a circuit in remapping a memory address, according to an embodiment.

[0017] FIG. 10 illustrates a block diagram of an example machine with which, in which, or by which any one or more of the techniques (e.g., methodologies) discussed herein can be implemented.

## DETAILED DESCRIPTION

[0018] After a memory device has been packaged, it is typically externally accessible only through nodes (e.g., contacts, pins, etc.) of the package, which can make repair of the memory device more difficult than prior to packaging. In addition, some defective memory cells might be identified (e.g., detected) after the memory device has been packaged and assembled into a larger unit, such as a dual-inline memory module (DIMM), flash memory card, smart phone, tablet computer, or the like. It is desirable to be able to repair memory devices after packaging, such as to improve overall yield and to reduce cost.

[0019] Memory devices may include groups (e.g., rows) of memory cells that can be used to replace defective groups (e.g., rows) of memory cells. In one such device, address data corresponding to a defective group of memory cells (hereinafter “defective address data”) can be stored in programmable elements, such as volatile or non-volatile memory. The programmable elements can be used to remap the address of the defective group of memory cells to another group of memory cells (e.g., a group of “redundant” memory cells) that is functional. Address data corresponding to an address to be accessed can be latched in response to an Activate command. The device can then check if the latched address data matches the defective address data by comparing the latched address data with the defective address data. A row decoder, for example, can cause a selected row of memory cells to be activated to access data associated with the latched address data. The selected row can be a row of redundant memory cells when the latched address data matches the defective address data stored by the programmable elements.

[0020] SPPR features can repair defective memory cells post-packaging. Soft post package repair can refer to a non-persistent method of post package repair. In soft post package repair, defective address data can be stored in volatile memory of the memory device after the memory device is packaged. The defective address data can, for example, correspond to a group of memory cells that were identified as defective post packaging. In some cases, the group of memory cells identified as defective post packaging could be a group of redundant memory cells to which an address has been previously remapped. In such cases, other defective address data can already be stored in programmable elements, such as antifuses, so that memory cells associated with the other defective address data are not accessed. The defective address data can be stored as part of a power-up sequence of a memory device, for example. The defective address data can be stored in volatile memory until the memory device is powered down. A storage element comprising volatile memory, such as latches, registers, and/or flip-flops, can store the defective address data and a decoder can

map the defective address to another group of memory cells. The other group of memory cells can be a group of redundant memory cells (e.g., a column or row of redundant memory cells) that are dedicated to soft post package repair. For instance, the defective address can be repeatedly mapped to the other group of memory cells.

[0021] Soft post package repair can be controlled by a controller that is external to the memory device. The controller can correspond to a memory controller, to test equipment, or the like. The controller can repair a packaged memory device by providing signals to nodes of the packaged memory device, in which defective address data is stored in a volatile manner. For example, the controller can provide signals on pre-existing Activate, Data Write, Data, and/or like nodes of a packaged DRAM device. In certain embodiments, the controller can redundantly store the defective address data in non-volatile memory stored outside of the packaged memory device that is being repaired for future retrieval. According to an embodiment, an apparatus can include a first memory device and a second memory device and the controller can be configured to retrieve defective address data from non-volatile memory of the second memory device and provide the retrieved defective address data to the first memory device. The first and second memory devices can be included in a packaged unit, such as a dual inline memory module or a memory card.

[0022] In an embodiment, soft post package repair can repair a memory device by remapping a defective address that had been previously remapped to a defective group of redundant memory cells. For example, a group of redundant memory cells to which an address has already been remapped could itself become defective post packaging. In one such embodiment, the defective address can be remapped to a different group of redundant memory cells.

[0023] A memory device that can perform soft post package repair can include memory cells in a package, volatile memory configured to store defective address data in response to entering a soft post-package repair mode, a match logic circuit, and a decoder. The match logic circuit can generate a match signal indicating whether address data corresponding to an address of the memory cells to be accessed matches the defective address data stored in the volatile memory. The decoder can select a first group of the memory cells to be accessed instead of a second group of the memory cells in response to the match signal indicating that the address data corresponding to the address to be accessed matches the defective address data stored in the volatile memory. The second group of the memory cells can correspond to a replacement address associated with other defective address data stored in non-volatile memory of the apparatus.

[0024] In the context of this document, a group of commonly coupled memory cells in a first general direction can correspond to, for example, a “row” of memory cells (which is also sometimes referred to herein as a “row of memory”). While a group of commonly coupled memory cells in another direction, typically a perpendicular direction can correspond to a “column” of memory cells. Repairing a row of memory can refer to, for example, reassigning the address previously assigned to a defective row of memory to another row of memory. “Programming,” “enabling,” and “disabling” a redundant row can mean, for example, programming, enabling, or disabling a group of redundant memory cells. A logic signal name ending with a trailing “F” can denote an active-low signal. However, it will be appreciated that the various embodiments of this document can include alternative active-high and/or active-low logic signals or logically equivalent circuits.

[0025] Within a group of cells (e.g., a row or column of memory cells), subgroups may be referred to as a “page” of cells, such that each group comprises multiple pages. The soft post package repair may be performed at the group or page level, such that one entry stored in the non-volatile memory of the apparatus may indicate that a defective address is for an entire group of memory cells and another entry stored in the non-volatile memory of the apparatus may indicate that a defective address is for a single page of memory cells. In some example embodiments, each group of cells comprises the same number of pages of cells.

[0026] In some memory devices, data in an entire row is available for access in a single activate

cycle. Thus, a single read request may be responded to with a single response comprising all of the data in a row identified by an address included in the read request. In other memory devices, less than all data in a row is available for access in a single activate cycle. The data that is available for access in a single activate cycle is referred to herein as a “page.” Accordingly, for some memory devices, each row includes a single page while in other memory devices, each row includes multiple pages. As discussed above, the row size is determined by the physical layout of the memory components (i.e., the number of commonly addressable memory cells in a first general direction). The page size may be determined by other design constraints (e.g., the number of connection pins on a memory sub-system, which may be lower than the row size).

[0027] The memory device may comprise prime rows and redundant rows. Initially, each prime row is associated with one or more unique memory addresses of the manufactured memory, and the redundant rows are unused. Accordingly, a memory access command that identifies a memory address will (initially) access a prime row. When defects in prime rows are detected, the defective pages or rows are retired from service and a redundant row is used in its place. Thus, the entire address range of the memory device may continue to be used even though a prime row associated with an address has become defective, extending the useful life of the memory device.

[0028] The redundant rows of the memory device may be pre-allocated (e.g., before or when the device is fabricated) as row-based redundant rows or page-based redundant rows. Alternatively, the redundant rows may be configurable as row-based or page-based redundant rows post packaging. Row-based redundant rows and page-based redundant rows may be physically identical but used differently for post-package repair. A row-based redundant row is used in place of an entire prime row. Each page of a page-based redundant row may be used in place of an individual page of a prime row. Row-based redundant rows may be used on a row-by-row basis or to repair multiple rows at a time.

[0029] Thus, when a single page of a prime row becomes defective, a page of a page-based redundant row may be used in place of the defective page while the remainder of the prime row continues to be used. By contrast, when a single page of a prime row becomes defective, the entirety of a row-based redundant row may be used in place of the entirety of the prime row, causing the non-defective portions of the prime row to cease being used. Additionally, compared with solutions using page-based redundant rows, either more row-based redundant rows are allocated or fewer page defects can be repaired using the same number of redundant rows. For example, if there are eight pages per row, a single page-based redundant row can be used to substitute pages for eight defective pages but eight row-based redundant rows would be used when the eight defective pages are in different prime rows.

[0030] The waste of non-defective data is increased when the row-based redundant rows are used to perform multiple-row repairs in which a single repair affects multiple (e.g., two, four, or eight) prime rows. By comparison with solutions using single-row repairs, multiple-row repairs reduce the number of address bits used to determine if a redundant row is being used and what the address of the redundant row is, but a single defective page in a single prime row can cause multiple non-defective rows to stop being used. Accordingly, compared with solutions using page-based redundant rows or single-row-based redundant rows, either more multiple-row-based redundant rows are allocated or fewer page defects can be repaired.

[0031] In some example embodiments, multiple-row-based redundant rows are used to repair defective pages or rows identified during the manufacturing process and a combination of single-row-based redundant rows and page-based redundant rows used to repair defective pages or rows identified after fabrication is complete.

[0032] FIG. 1 is a block diagram illustrating a memory bank of a memory device **100** configured to remap memory addresses using row-based repairs and page-based repairs, according to an embodiment. The illustrated memory device **100** includes a control logic circuit **110**, a storage element **112**, a match logic circuit **114**, a row enable circuit **116**, a row decoder **118**, a memory

array **120** comprising data rows **122**, row-based redundant rows **124**, and page-based redundant rows **128**, and a programmable element bank **126**. The memory device **100** can include a mode register, and a soft post package repair mode can be entered in response to a change in a value stored in the mode register. The memory device **100** can include more or fewer elements than illustrated in FIG. 1.

[0033] The control logic circuit **110** can receive a soft post package repair signal SPPR and an activate signal Activate as inputs and generate a pulse signal Pulse and a soft post package repair pulse SPPR Pulse. Both of these pulse signals can be asserted for a sufficient amount of time for volatile memory elements, such as latches, registers, SRAM, or the like, to capture data. Then the control logic circuit **110** can de-assert these pulse signals after the data has been captured. The soft post package repair signal SPPR can be asserted to enter a soft post package repair mode of operation. When the memory device **100** is operating in the soft post package repair mode, the activate signal Activate can be asserted when defective address data is provided to the memory device **100**. The control logic circuit **110** can assert the soft post package repair pulse SPPR Pulse in response to the activate signal Activate being asserted when the memory device **100** operates in the soft post package repair mode. The control logic circuit **110** can assert the pulse signal Pulse in response to the activate signal Activate being asserted when the memory device **100** operates in a mode of operation other than the soft post package repair mode.

[0034] The storage element **112** can receive address data Address [N:O] and store the address data Address [N:O] in volatile memory elements. The storage element **112** can comprise any suitable volatile memory elements (e.g., SRAM) to store the address data Address [N:O]. The storage element **112** can include a first group of memory elements and a second group of memory elements. These groups of memory elements can be logically separate from each other. In some instances, these groups of memory elements can also be physically separate from each other. The first group of memory elements can store defective address data. The address data Address [N:O] can be stored in the first group of memory elements as defective address data when the soft post package repair pulse signal SPPR Pulse is asserted. The second group of memory elements can store address data corresponding to an address of the memory array **120** to be accessed (e.g., read or programmed). The address data Address [N:O] can be stored in the second group of memory elements when the pulse signal Pulse is asserted.

[0035] The match logic circuit **114** can compare address data from the first group of memory elements SPPR Addr\_Lat [N:O] with address data from the second group of memory elements Addr\_Lat [N:O]. The match logic circuit **114** can generate the soft post package repair match signal SPPR Match in response to (e.g., based at least in part on) the comparison. The soft post package repair match signal SPPR Match can be asserted to indicate that the address to be accessed in the memory array **120** matches a defective address stored in the storage element **112**. When the address to be accessed does not match a defective address stored in the storage element, the soft post package repair match signal SPPR Match can be de-asserted. The match logic circuit can also de-assert the soft post package repair match signal SPPR Match in response to a reset signal Reset.

[0036] The row enable circuit **116** can include circuitry configured to control activating a row of the memory array **120**. The row enable circuit **116** can receive the activate signal Activate and provide a row enable signal Row Enable to the row decoder **118**. The row enable circuit **116** can provide a delay in enabling the row decoder **118** after the activate signal Activate is asserted. This can enable the row decoder **118** to select a row of the memory array **120** when address data and match signals are ready to provide a selected row address of the memory array **120**. The selected row address can be an address remapped to a redundant row of the row-based redundant rows **124** to repair a defective data row of the data rows **122** or a defective redundant row of the row-based redundant rows **124**. The selected row address may also be an address remapped to a redundant row of the page-based redundant rows **128** to repair a defective data row of the data rows **122** or a defective redundant row of the row-based redundant rows **124**. Each of the row-based redundant

rows **124**, when in use, is used in place of an entire row of the data rows **122**. By contrast, the page-based redundant rows **128** are used a page at a time. Thus, a page-based redundant row **128** may be selected by the row decoder **118** based not only on the row identified by the address, but by the page as well.

[0037] The row decoder **118** can decode the address to be accessed to select a row of memory cells in the memory array **120** to which the address to be accessed is mapped. The selected row can be a data row of the data rows **122** when the address to be accessed is not known to be defective. For example, when the row enable signal Row Enable is asserted and the address to be accessed does not match a defective address stored in either storage element **112** or programmable element bank **126**, a prime row signal Prime Row can be asserted to activate a row of the data rows **122** mapped to the address to be accessed as the selected row. The row decoder **118** can prevent a defective row from being activated in response to the soft post package repair match signal SPPR Match from the match logic circuit **114** and/or a redundant match signal Redundant Match from the programmable element bank **126**.

[0038] When the address to be accessed matches a defective address stored in the storage element **112** and/or a defective address stored in the programmable element bank **126**, the programmable element bank **126** identifies the address of the redundant row to be used in place of the defective row. In response to the redundant match signal, the row decoder **118** decodes the redundant address instead of the latched address. The row decoder **118** can decode the address to be accessed to select a redundant row of the row-based redundant rows **124** or the page-based redundant rows **128**. A redundant row signal Row-Based Redundant Row can be asserted to activate a row of the row-based redundant rows **124** mapped to the address to be accessed as the selected row. In response to the soft post package repair match signal SPPR Match being asserted, a different row of the row-based redundant rows **124** can be activated instead of a row of the row-based redundant rows **124** that would have been activated if only the redundant match signal Redundant Match had been asserted. Alternatively, a redundant row signal Page-Based Redundant Row can be asserted to activate a row of the page-based redundant rows **128** mapped to the address to be accessed as the selected row.

[0039] For example, in response to the redundant match signal Redundant Match being asserted, the row decoder **118** can decode redundant row address data Redundant Section <M:O>, provided by the programmable element bank **126** as corresponding to the address to be accessed, to select a redundant row of the row-based redundant rows **124** to which a defective address has been originally remapped. In response to the soft post package repair match signal SPPR Match indicating that the address to be accessed matches a defective address stored in the storage element **112**, the row decoder **118** can select a different redundant row of the row-based redundant rows **124** instead, such as in a case where the redundant row to which the defective address had been originally remapped has later been identified as defective. Accordingly, storing the defective address data in the storage element **112** can be used to prevent a defective row of the row-based redundant rows **124** from being selected.

[0040] The memory array **120** can include volatile or nonvolatile memory cells. Some examples of memory cells that can be implemented in the memory array **120** include DRAM cells, SRAM cells, flash memory cells, such as NAND flash memory cells, phase change memory cells, ferroelectric random access memory (FeRAM), and the like; and the memory array may include hybrid memory devices implementing including multiple memory technologies. The programmable element bank **126** can store defective address data in the memory array **120** using programmable elements. The programmable element bank **126** can compare address data corresponding to an address to be accessed Address\_Lat <N:O> with the defective address data stored by the programmable elements to determine whether the address to be accessed corresponds to a defective address of the array **120**. The programmable element bank **126** can generate a redundant match signal indicative of whether the address to be accessed matches a defective address of the memory array **120**. The

programmable element bank **126** can provide redundant address data Redundant Address <N:0> and Redundant Section <M:0> used to select a row of the row-based redundant rows **124** or the page-based redundant rows **128** to which the defective address has been remapped.

[0041] FIG. **1** illustrates one bank of a memory device **100**. In certain embodiments, some or all of the circuits illustrated in FIG. **1** can be replicated in two or more banks of the memory device **100**. Accordingly, the soft post package repair described with reference to FIG. **1** can be a per bank solution. In one embodiment, multiple banks of the memory device **100** can simultaneously perform soft post package repair. Alternatively or additionally, two or more banks of the memory device **100** can sequentially perform soft post package repair.

[0042] FIG. **2** is a flow chart showing operations of a method **200** performed by a circuit in remapping memory addresses, according to an embodiment. The method **200** can be implemented by the memory device **100** of FIG. **1**, for example. Any combination of the features of the method **200** can be embodied in code stored in a non-transitory computer-readable storage. When executed, the non-transitory computer-readable storage can cause some or all of any of the method **200** to be performed. It will be understood that any of the methods discussed herein may include greater or fewer operations and that the operations may be performed in any order, as appropriate.

[0043] The method **200** can perform soft post package repair. In one embodiment, the method **200** is performed as part of a power-up sequence of a memory device and/or in response to an instruction received from a device outside the package, such as from a memory controller or from test equipment. At block **202**, the soft post package repair mode can be entered. The soft post package repair mode can be entered in response to receiving a soft post package repair command at an externally accessible node of a package of a memory device, such as the memory device **100**. Prior to entering the soft post package repair mode, operations of other post package repair modes can be performed.

[0044] While operating in the soft post package repair mode, an activate command can be received at block **204**. The activate command can be followed by a precharge. Defective address data can be provided with the activate command. The defective address might be currently mapped to a group of “regular” memory cells or might have already been remapped to a group of redundant memory cells. The defective address data can be stored in volatile memory of a storage element, such as a latch or a register, at block **206**. After the defective address data is stored, the next one of multiple latch signals will toggle to accept the address presented into the next unused SPPR page storage latch. Thus, more than one defective address can be stored in the storage element during the soft post package repair mode such that multiple defective portions of the memory array can be repaired by soft post package repair. After the defective address data is stored in the storage element, soft post package repair mode can be exited at block **208**.

[0045] During operation of the memory device, memory access operations can be performed in response to inputs received at one or more externally accessible nodes of the package of the memory device. The memory access operations can include read operations and/or programming (write) operations. As part of the memory access operations, an activate command can be received. The activate command can be received with an address to be accessed at block **210**. The address data corresponding to the address to be accessed can also be stored in volatile memory of a storage element at block **210**.

[0046] The address to be accessed can be compared to the defective address (e.g., multiple defective addresses) to determine whether there is a soft post package repair match at block **212**. When the address to be accessed matches a defective address stored in the storage element, a soft post package repair match is present. The defective address can be remapped to a group of functional memory cells of the memory array at block **214**. A decoder can perform such remapping. The decoder can also override a replacement address associated with other defective address data previously stored in non-volatile memory, in which the replacement address corresponds to a different (e.g., now defective) group of memory cells that were previously mapped to the address to



be accessed. The replacement address can be mapped to a location of the non-volatile memory that stores the other defective address data. For instance, replacement address data can be generated by a decoder based on the location of the non-volatile memory that stores the other defective address data. Alternatively, the replacement address can be stored in the non-volatile memory. The process can then continue at block **210**.

[0047] On the other hand, when the address to be accessed does not match a defective address stored in the storage element at block **212**, a soft post package repair match is not present. In that case, the address to be accessed can be decoded by a decoder at block **216**. The decoder can select a group of memory cells of the memory array mapped to the address to be accessed. The selected group of memory cells of the memory array can be either a group of regular memory cells or a group of redundant memory cells of the memory array. The process can then continue at block **210**.

[0048] FIG. **3** is a block diagram **300** illustrating page-based repairs, according to an embodiment. Page-based redundant row **310** comprises 128 pages of memory (e.g., FeRAM) that can be individually used in place of bad pages in prime rows. In this example, each row is 8192 bits and each page is 64 bits. Among the 128 pages of memory are pages **320** and **330** (indicated as Page 1 and Page 4 in FIG. **3**). Prime row **340** has 127 good pages as well as bad page **350**. Prime row **360** has 127 good pages as well as bad page **370**.

[0049] The address of bad pages **350** and **360** are stored so that when a memory access request is received that includes an address corresponding to a bad page, the memory of a page of the page-based redundant row **310** is accessed instead. In the example of FIG. **3**, the page **320** is accessed instead of the bad page **350** and the page **330** is accessed instead of the bad page **370**.

[0050] By contrast, in a row-based repair, an entire redundant row would be used in place of each of the prime rows **340** and **360**. This would waste the 127 good pages in each of the prime rows **340** and **360**, which would cease being used just because a single page went bad. Furthermore, instead of using one page-based redundant row to repair pages in up to 128 different prime rows, two row-based redundant rows would be used just to repair the two bad pages shown in FIG. **3**.

[0051] FIG. **4** is a block diagram illustrating an example row decoder **400**, according to an embodiment. The row decoder **400** includes an address decoder **405**, an SPPR circuit **410**, a redundant row decoder **415**, AND gates **420**, **435**, **445**, **450**, and **455**, OR gates **425** and **440**, and NOT gates **430** and **460**. The row decoder **400** receives an address (AddLat<16:0>), a row enable, a redundant section (Redundant Section<3:0>), a redundant address (Redundant Address <16:0>) and a Redundant Match as inputs and generates an output for each row in a memory array. The output for a row is set when the row is being accessed and cleared otherwise. Thus, the row decoder **400** may be configured to accept a multi-bit address (e.g., a 17-bit address) and decode the address to identify a single row of the memory array. With reference to FIG. **1**, the row decoder **118** may receive the address from the storage element **112**, the row enable from the row enable circuit **116**, and the Redundant Match, the redundant section, and the redundant address from the programmable element bank **126**. The outputs may be provided to the memory array **120**.

[0052] The address decoder **405** receives the address input and generates multiple outputs, one for each prime row. One of the outputs is set, indicating that the corresponding prime row is the row for the input address. The remaining outputs are cleared, indicating that the corresponding prime row is not being accessed. The output for each prime row may be implemented using multiple lines, one for each page of the prime row. In the example of FIG. **4**, four pages per row are used. Accordingly, four outputs (shown as a four-bit value, <3:0>) are provided for each prime row. Only one output is set, indicating the specific page of the specific row to be accessed.

[0053] The AND gate **420** is replicated for each output of the address decoder **405**. For all prime rows that do not correspond to the input address, the input to the AND gate **420** from the address decoder **405** is cleared, so the output from the AND gate **420** is also cleared, and the corresponding prime row is not accessed. For the prime row that corresponds to the input address, the prime row is selected, the row enable input is set, and the output of the NOT gate **430** is set. The output of the

NOT gate **430** is set if and only if the SPPR Match signal and the Redundant Match signal are both cleared. Thus, if either a Redundant Match or SPPR Match is found, no prime row will be selected, and when neither a Redundant Match nor an SPPR Match is found, the prime row determined by the address decoder **405** will be selected.

[0054] The SPPR circuit **410** determines whether the input address is for an SPPR. If so, the SPPR circuit **410** indicates this by setting the SPPR Match output. Otherwise, the SPPR Match output is cleared.

[0055] Each bit of the redundant section input is provided to a different one of the logic gates **440-455**. Each of the logic gates **440-455** takes SPPR Match or its inverse as the second input.

Combined, the logic gates **440-455** generate a modified redundant section output that is provided to the redundant row decoder **415**.

[0056] If there is a redundant match or an SPPR match, the output of the OR gate **425** is set. If the row enable input is also set, the output of the AND gate **435** is set, enabling the redundant row decoder **415**. When enabled, based on the modified redundant section and the redundant address, the redundant row decoder **415** indicates whether a redundant row or SPPR redundant row should be used and selects the row. When disabled, the redundant row decoder **415** indicates that no redundant row should be used.

[0057] If the address decoder **405** determines that the input address is valid, the row enable signal is received, and there is neither a redundant match nor an SPPR match (causing the output of the OR gate **425** to be cleared and the output of the NOT gate **430** to be set) and the output of the AND gate **420** is set, indicating that a prime row should be used.

[0058] FIG. 5 is a block diagram illustrating an example row decoder **500**, according to an embodiment. The row decoder **500** includes the address decoder **405**, an SPPR circuit **510**, a redundant row decoder **515**, AND gates **420**, **435**, **445**, **450**, and **455**, OR gates **425** and **440**, and NOT gates **430** and **460**. The row decoder **500** receives an address (AddLat<16:0>), a row enable, a redundant section (Redundant Section <3:0>), a redundant address (Redundant Address <16:0>) and a Redundant Match as inputs and generates a prime row selection, a redundant row selection, and an SPPR redundant row selection as outputs. With reference to FIG. 1, the row decoder **118** may receive the address from the storage element **112**, the row enable from the row enable circuit **116**, and the Redundant Match, the redundant section, and the redundant address from the programmable element bank **126**. The outputs may be provided to the memory array **120**.

[0059] The address decoder **405**, the AND gates **420**, **435**, **445**, **450**, and **455**, the OR gates **425** and **440**, and the NOT gates **430** and **460** operate as discussed above with respect to FIG. 4. The SPPR circuit **510** is modified from the SPPR circuit **410** to include an SMPage <3:0> output. The SMPage <3:0> output allows for selection of an individual page of the selected row.

[0060] Thus, redundant row decoder **515** is enabled to select a row based on the redundant address and a page of the row based on the SMPage <3:0> input. By contrast, the redundant row decoder **415** of FIG. 4 does not receive an SMPage <3:0> input and thus selects the page only based on the redundant address.

[0061] FIG. 6 is a block diagram illustrating the SPPR circuit **510**, according to an embodiment. The SPPR circuit **510** receives an address via an address bus as an input and generates an SPPR Match signal and an SMPage signal as outputs. The input address is an address for memory to be accessed. The SPPR Match output is set if the prime row for the address should be ignored and a redundant row used instead. In that case, the SMPage signal indicates which page of the redundant row (e.g., a redundant row identified by the redundant address input to the redundant row decoder **515** of FIG. 5) is to be used.

[0062] The SPPR circuit **510** comprises latches **610**, **620**, **630**, and **640**, XNOR gates **650**, **660**, **670**, and **680**, and OR gate **690**. The latches **610-640** store data that indicates whether, for an input address, a particular page has been repaired. For example, the address bus may provide a 17-bit address. The high 15 bits of the address may identify a row in a memory and the low 2 bits may

select from among four pages in each row. Each of the latches **610-640** represents a number of latches equal to the size of the address (e.g., 17 latches for a memory using 17-bit addresses). Similarly, each of the XNOR gates **650-680** represents a number of XNOR gates equal to the size of the address. An XNOR gate outputs a 1 when the two inputs are the same and a 0 if they are different. Thus, an XNOR gate that receives one bit of a stored address and one bit of an input address will output a 1 if the bits match. The outputs of the multiple XNOR gates are ANDed together to generate a 1-bit output that indicates whether the input address matches the latched address. In the example of FIG. 5, latches are available to store four addresses, allowing for four any-to-any soft page repairs. In other example embodiments, more or fewer any-to-any soft page repairs may be available, and a corresponding number of latches are used.

[0063] The 17 bits of the address may be provided as inputs to each of the latches **610-640**. Each latch determines if the corresponding row has been repaired for a specific page (e.g., latch **610** for page 0, latch **620** for page 1, latch **630** for page 2, latch **640** for page 3). Thus, for any given address, the signal shown from the address bus to the XNOR gates **650-680** is compared to latched repair addresses from latches **610-640** by XNOR gates **650-680** to determine if there is a match.

[0064] When the output of one of the XNOR gates **650-680** is set, this indicates that the corresponding page is being used for an SPPR repair. The outputs of the XNOR gates **650-680**, together, are the SMPage <3:0> output, indicating which page is selected for the SPPR repair. If any bits of the SMPage <3:0> output are set, the OR gate **690** causes the SPPR Match output to be set.

[0065] To record an address in one of the latches **610-640**, a latch signal is provided to the SPPR circuit **510** at the same time an address is provided. The SPPR circuit **510** modifies which of the latches **610-640** is coupled to the latch signal after each time one of the latches **610-640** stores an address. Thus, the latched addresses are recorded in a particular one of the latches **610-640** without being recorded in the others.

[0066] FIG. 7 is a block diagram illustrating the redundant row decoder **415**, according to an embodiment. The redundant row decoder **415** comprises AND gates **705, 710, 715, 720, 730, 735, 740, and 745**, NOT gate **725**, and OR gate **750**. The redundant row decoder **515** receives an SPPR Match signal, an SMPage <3:0> signal, a Redundant Address <16:0> signal, a Redundant Section <3:0> signal, two address bits (X and Y) that determine a page, and the inverse of the two address bits (XF and YF) as inputs. The redundant row decoder **515** generates a Row <3:0> output that indicates the page of the redundant row to be read.

[0067] If SPPR Match is set, the output of the NOT gate **725** is cleared and the output of all four AND gates **730-745** is cleared. Then the output of the bitwise OR gate **750** is all zeros, regardless of the value of SMPage <3:0>. As a result, the OR gate **750** ORs the SMPage <3:0> signal with zeros and the Page <3:0> value is the same as the SMPage <3:0> signal. Thus, when the SPPR circuit **410** determines that there is an SPPR Match and identifies the corresponding SMPage<3:0>, the redundant row decoder **415** does not override the determined page based on the redundant address provided by the programmable element bank **126**.

[0068] If SPPR Match is cleared, the AND gates **705-720** and **730-745** act as a two-bit decoder, causing a different one of the four AND gates **730-745** to have an output value of one for each of the four possible combinations of X and Y. The decoded row is bitwisely ORed with the SMPage <3:0> by the OR gate **750** and provided as the Page <3:0> input to the row decoder **755**. SMPage <3:0> is all zeros if SPPR Match is cleared, so the Page <3:0> input is just the decoded row.

[0069] The row decoder **755** has many outputs, one for each redundant row of the memory array **120**. At most one of the outputs is set, indicating that the corresponding row is to be read. The outputs may comprise multiple bits (e.g., one for each page of the corresponding row). For simplicity, only the output corresponding to the selected row is shown in FIG. 7.

[0070] The row decoder **755** selects a row based on the Redundant Address <16:0> input and the Redundant Section <3:0> input. The page of the row is selected based on the Page <3:0> input. The

output signal for the selected row and page is set and all other outputs are cleared.

[0071] FIG. 8 is a flow chart showing operations of a method **800** performed by a circuit in remapping a memory address, according to an embodiment. The method **800** comprises operations **810**, **820**, and **830**. By way of example and not limitation, the method **800** is described as being performed by the memory device **100** of FIG. 1.

[0072] In operation **810**, the memory device **100** receives a memory access command comprising an address. For example, a read command may be received along with an address on the Address <N:0>signal of FIG. 1.

[0073] The memory device **100**, in operation **820**, determines, based on the address, that a first page for the address in a first row of the memory device **100** is defective. For example, the storage element **112** may latch the input address and provide it as an input to the programmable element bank **126**. Based on the address, the programmable element bank **126** accesses a storage element that indicates whether the address is for a row having one or more defective pages. If one or more pages of the row are defective, the programmable element bank generates an output that indicates this (e.g., the Redundant Match signal of FIG. 1) and an address for a row that is being used in place of the first row (e.g., the Redundant Address <N:0>signal of FIG. 1).

[0074] Based on the first page being defective, the memory device **100** accesses a second page from a page-based redundant row of the memory device **100** (operation **830**). For example, the redundant row decoder **415** of FIG. 4 may use an output from the SPPR circuit **410** along with the Redundant Address <N:0>signal from the programmable element bank **126** to select a page of a row-based redundant row. The memory device **100** accesses the selected page from the memory array **120** (e.g., one of the page-based redundant rows **128**).

[0075] The method **800** may be repeated any number of times for different addresses or for the same address at different times. Operation **820** recites determining that a page in a row is defective based on the address. In the alternative, the memory device **100** may, based on the address, determine that the page in the row is not defective. In that case, based on the page of the row not being defective, the memory device **100** accesses a page from the row. For example, the programmable element bank **126** may store addresses for defective rows. If the address received by the memory device is not stored in the programmable element bank **126**, the prime row for the address may be accessed.

[0076] FIG. 9 is a flow chart showing operations of a method **900** performed by a circuit in remapping memory address, according to an embodiment. The method **900** comprises operations **910**, **920**, **930**, **940**, **950**, **960**, and **970**. By way of example and not limitation, the method **900** is described as being performed by the memory device **100** of FIG. 1.

[0077] In operation **910**, the memory device **100** receives a memory access command comprising an address. In operation **920**, the memory device **100** determines if a row for the address is defective. For example, the first two bits of the address may identify a section of the memory array **120**, the next ten bits of the address may identify a row of the memory array **120** within the section, and the last three bits of the address may identify a page within the row. A lookup table in the programmable element bank **126** may store the section and row portions of addresses for which the corresponding data rows are at least partially defective. Thus, if the received address is stored in the programmable element bank **126**, the memory device **100** determines that the row for the address is defective. If the row for the address is not defective, the memory device **100** accesses the row (operation **930**).

[0078] If the row for the address is determined to be defective in operation **920**, the memory device **100** further determines if the entire row for the address is defective or if the row is only partially defective (operation **940**). For example, the SPPR Circuit **410** may store the complete addresses for defective pages. If any pages for the row are stored in the SPPR Circuit **410**, the row is only partially defective. If the row is fully defective, a row-based redundant row for the address is accessed in operation **960**. For example, the address for the row-based redundant row may be

stored in the programmable element bank **126** and provided from the programmable element bank **126** to the redundant row decoder **415**. The redundant row decoder **415** decodes the address and identifies the corresponding row-based redundant row.

[0079] If the row is only partially defective and the page for the address is not defective, the page is accessed from the row for the address (operation **930**). Thus, the non-defective portions of a partially defective row remain in use, increasing the lifespan of the memory device **100**.

[0080] If the row is only partially defective and the page for the address is defective, a page of a page-based redundant row is accessed (operation **970**). For example, the SPPR circuit **410** may provide an address to the redundant row decoder **415**. The redundant row decoder **415** decodes the address and identifies the corresponding page-based redundant row.

[0081] The same page-based redundant row may be used in place of defective pages in different prime rows. Conversely, a single prime row may have multiple pages replaced using the same or different page-based redundant rows.

[0082] The method **900** may be used to substitute pages in a page-based redundant row for pages in a row-based redundant row. For example, if an entire prime row is found to be defective, the portion of an address that identifies the row may be stored in the programmable element bank **126** along with an address for a row-based redundant row to be used in substitution for the prime row. At a later time, one of the pages in the row-based redundant row may become defective. Thereafter, the address for the corresponding page in the prime row is stored in the SPPR Circuit **410**. As a result, when an incoming memory access command is received that addresses the doubly defective page, the programmable element bank **126** will identify the row-based redundant row that replaces the entire row and the SPPR Circuit **410** will identify the page of the page-based redundant row that replaces the page of the prime row (and of the row-based redundant row). Thus, the redundant row decoder **415** will receive a signal from the programmable element bank **126** to use a row-based redundant row and a signal from the SPPR Circuit **410** to use a page-based redundant row. The signal from the SPPR Circuit **410** overrides the signal from the programmable element bank **126** and the page-based redundant row is accessed.

[0083] FIG. **10** illustrates a block diagram of an example machine **1000** with which, in which, or by which any one or more of the techniques (e.g., methodologies) discussed herein can be implemented. Examples, as described herein, can include, or can operate by, logic or a number of components, or mechanisms in the machine **1000**. Circuitry (e.g., processing circuitry) is a collection of circuits implemented in tangible entities of the machine **1000** that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership can be flexible over time. Circuitries include members that can, alone or in combination, perform specified operations when operating. In an example, hardware of the circuitry can be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry can include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a machine-readable medium physically modified (e.g., magnetically, electrically, moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation. In connecting the physical components, the underlying electrical properties of a hardware constituent are changed, for example, from an insulator to a conductor or vice versa. The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, in an example, the machine-readable medium elements are part of the circuitry or are communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components can be used in more than one member of more than one circuitry. For example, under operation, execution units can be used in a first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry at a different time. Additional examples of these components with respect to the machine **1000**.

[0084] In alternative embodiments, the machine **1000** can operate as a standalone device or can be connected (e.g., networked) to other machines. In a networked deployment, the machine **1000** can operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the machine **1000** can act as a peer machine in peer-to-peer (P2P) (or other distributed) network environment. The machine **1000** can be a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a mobile telephone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), other computer cluster configurations.

[0085] The machine **1000** (e.g., computer system) can include a hardware processor **1002** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **1004**, a static memory **1006** (e.g., memory or storage for firmware, microcode, a basic-input-output (BIOS), unified extensible firmware interface (UEFI), etc.), and mass storage device **1008** (e.g., hard drives, tape drives, flash storage, or other block devices) some or all of which can communicate with each other via an interlink **1030** (e.g., bus). The machine **1000** can further include a display device **1010**, an alphanumeric input device **1012** (e.g., a keyboard), and a user interface (UI) navigation device **1014** (e.g., a mouse). In an example, the display device **1010**, the input device **1012**, and the UI navigation device **1014** can be a touch screen display. The machine **1000** can additionally include a signal generation device **1018** (e.g., a speaker), a network interface device **1020**, and one or more sensor(s) **1016**, such as a global positioning system (GPS) sensor, compass, accelerometer, or other sensor. The machine **1000** can include an output controller **1028**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate or control one or more peripheral devices (e.g., a printer, card reader, etc.).

[0086] Registers of the hardware processor **1002**, the main memory **1004**, the static memory **1006**, or the mass storage device **1008** can be, or include, a machine-readable media **1022** on which is stored one or more sets of data structures or instructions **1024** (e.g., software) embodying or used by any one or more of the techniques or functions described herein. The instructions **1024** can also reside, completely or at least partially, within any of registers of the hardware processor **1002**, the main memory **1004**, the static memory **1006**, or the mass storage device **1008** during execution thereof by the machine **1000**. In an example, one or any combination of the hardware processor **1002**, the main memory **1004**, the static memory **1006**, or the mass storage device **1008** can constitute the machine-readable media **1022**. While the machine-readable media **1022** is illustrated as a single medium, the term “machine-readable medium” can include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) configured to store the one or more instructions **1024**.

[0087] The term “machine readable medium” can include any medium that is capable of storing, encoding, or carrying instructions for execution by the machine **1000** and that cause the machine **1000** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding, or carrying data structures used by or associated with such instructions. Non-limiting machine-readable medium examples can include solid-state memories, optical media, magnetic media, and signals (e.g., radio frequency signals, other photon-based signals, sound signals, etc.). In an example, a non-transitory machine-readable medium comprises a machine-readable medium with a plurality of particles having invariant (e.g., rest) mass, and thus are compositions of matter. Accordingly, non-transitory machine-readable media are machine readable media that do not include transitory propagating signals. Specific examples of non-transitory machine-readable media can include: non-volatile memory, such as semiconductor memory devices

(e.g., electrically programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks.

[0088] In an example, information stored or otherwise provided on the machine-readable media **1022** can be representative of the instructions **1024**, such as instructions **1024** themselves or a format from which the instructions **1024** can be derived. This format from which the instructions **1024** can be derived can include source code, encoded instructions (e.g., in compressed or encrypted form), packaged instructions (e.g., split into multiple packages), or the like. The information representative of the instructions **1024** in the machine-readable media **1022** can be processed by processing circuitry into the instructions to implement any of the operations discussed herein. For example, deriving the instructions **1024** from the information (e.g., processing by the processing circuitry) can include: compiling (e.g., from source code, object code, etc.), interpreting, loading, organizing (e.g., dynamically or statically linking), encoding, decoding, encrypting, unencrypting, packaging, unpackaging, or otherwise manipulating the information into the instructions **1024**.

[0089] In an example, the derivation of the instructions **1024** can include assembly, compilation, or interpretation of the information (e.g., by the processing circuitry) to create the instructions **1024** from some intermediate or preprocessed format provided by the machine-readable media **1022**. The information, when provided in multiple parts, can be combined, unpacked, and modified to create the instructions **1024**. For example, the information can be in multiple compressed source code packages (or object code, or binary executable code, etc.) on one or several remote servers. The source code packages can be encrypted when in transit over a network and decrypted, uncompressed, assembled (e.g., linked) if necessary, compiled or interpreted (e.g., into a library, stand-alone executable etc.) at a local machine, and executed by the local machine.

[0090] The instructions **1024** can be further transmitted or received over a communications network **1026** using a transmission medium via the network interface device **1020** utilizing any one of a number of transfer protocols (e.g., frame relay, internet protocol, transmission control protocol (TCP), user datagram protocol (UDP), hypertext transfer protocol (HTTP), etc.). Example communication networks can include a local area network (LAN), a wide area network (WAN), a packet data network (e.g., the Internet), mobile telephone networks (e.g., cellular networks), plain old telephone (POTS) networks, and wireless data networks (e.g., Institute of Electrical and Electronics Engineers (IEEE) 802.11 family of standards known as Wi-Fi®, IEEE 802.16 family of standards known as WiMax®, IEEE 802.15.4 family of standards, peer-to-peer (P2P) networks, among others. In an example, the network interface device **1020** can include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the network **1026**. In an example, the network interface device **1020** can include a plurality of antennas to wirelessly communicate using at least one of single-input multiple-output (SIMO), multiple-input multiple-output (MIMO), or multiple-input single-output (MISO) techniques. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding or carrying instructions for execution by the machine **1000**, and includes digital or analog communications signals or other intangible medium to facilitate communication of such software. A transmission medium is a machine readable medium.

[0091] To better illustrate the methods and apparatuses described herein, a non-limiting set of Example embodiments are set forth below as numerically identified Examples.

[0092] Example 1 is a memory device comprising: a memory array comprising: a plurality of prime rows, each prime row comprising a first number of pages, the first number being greater than one; and a set of redundant rows, each redundant row comprising the first number of pages; a memory controller configured to: receive a memory access command comprising an address; and based on the address: determine that a first page for the address in a prime row is defective; and access a second page from a redundant row.

[0093] In Example 2, the subject matter of Example 1, wherein the memory controller is further configured to: receive a second memory access command comprising a second address; and based on the second address: determine that a third page for the second address in the prime row is not defective; and access the third page from the prime row.

[0094] In Example 3, the subject matter of Examples 1-2, wherein the memory controller is further configured to: receive a second memory access command comprising a second address; and based on the second address: determine that a third page for the second address in a second prime row is defective; and access a fourth page from the redundant row.

[0095] In Example 4, the subject matter of Examples 1-3, wherein the memory controller is further configured to: receive a second memory access command comprising a second address; and based on the second address: determine that an entire second prime row is being substituted by a second redundant row; determine that a third page in the second redundant row is defective; determine that the third page in the second redundant row is being substituted by a fourth page in a third redundant row; and access the fourth page from the third redundant row.

[0096] In Example 5, the subject matter of Example 4 includes a logic circuit configured to assert a match signal when address data to be accessed matches defective address data; wherein the determining that the third page in the second redundant row is defective is based on the match signal being asserted.

[0097] In Example 6, the subject matter of Examples 1-5 includes a logic circuit configured to assert a match signal when address data to be accessed matches defective address data; wherein the determining that the first page for the address in the prime row is defective is based on the match signal being asserted.

[0098] In Example 7, the subject matter of Examples 1-6, wherein the memory array comprises a hybrid random access memory array.

[0099] Example 8 is a method comprising: receiving, by a memory controller, a memory access command comprising an address corresponding to a page of memory, a page being smaller than a row; and based on the address: determining that a first page for the address in a first row of a memory device is defective; and accessing a second page from a redundant row of the memory device.

[0100] In Example 9, the subject matter of Example 8 includes receiving a second memory access command comprising a second address; and based on the second address: determining that a third page for the second address in the first row is not defective; and accessing the third page from the first row.

[0101] In Example 10, the subject matter of Examples 8-9 includes receiving a second memory access command comprising a second address; and based on the second address: determining that a third page for the second address in a second prime row of the memory device is defective; and accessing a fourth page from the redundant row of the memory device.

[0102] In Example 11, the subject matter of Examples 8-10 includes receiving a second memory access command comprising a second address; and based on the second address: determining that an entire second prime row is being substituted by a second redundant row; determining that a third page in the second redundant row is defective; determining that the third page in the second redundant row is being substituted by a fourth page in a third redundant row; and accessing the fourth page from the third redundant row.

[0103] In Example 12, the subject matter of Example 11, wherein: the determining that the third page in the second redundant row is defective is based on a match signal being asserted by a logic circuit that is configured to assert the match signal when address data to be accessed matches defective address data.

[0104] In Example 13, the subject matter of Examples 8-12, wherein: the determining that the first page for the address in the first row is defective is based on a match signal being asserted by a logic circuit configured to assert the match signal when address data to be accessed matches defective



address data.

[0105] Example 14 is a method comprising: detecting, after packaging a memory device that comprises a memory array comprising prime rows and redundant rows and non-volatile memory, first defective address data corresponding to a first address of the memory array to be accessed that includes, at least one defective memory cell; and storing, after packaging the memory device, the first defective address data into the non-volatile memory in association with a page of a redundant row, wherein a page is smaller than a row.

[0106] In Example 15, the subject matter of Example 14 includes detecting, before packaging the memory device, second defective address data corresponding to a second address of the memory array to be accessed that includes at least one defective memory cell; and storing, before packaging the memory device, the second defective address data into the non-volatile memory in association with an entire second redundant row.

[0107] In Example 16, the subject matter of Example 15, wherein the storing of the second defective address data in association with the second redundant row is based on the detecting of the second address being defective before the packaging of the memory device.

[0108] In Example 17, the subject matter of Examples 14-16, wherein the storing of the first defective address data in association with the redundant row is based on the detecting of the first address being defective after the packaging of the memory device.

[0109] In Example 18, the subject matter of Examples 14-17 includes receiving, by a memory controller of the memory device, a memory access command corresponding to the first address; based on the stored first defective address data: determining that a first page for the first address in a prime row of the memory device is defective; and accessing a second page from the redundant row of the memory device.

[0110] In Example 19, the subject matter of Example 18 includes receiving a second memory access command comprising a second address; determining that a third page for the second address in the prime row is not defective; and accessing the third page from the prime row.

[0111] In Example 20, the subject matter of Examples 18-19 includes determining that the first page for the first address in the prime row is defective based on a match signal being asserted by a logic circuit configured to assert the match signal when address data to be accessed matches defective address data.

[0112] Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.

[0113] Example 22 is an apparatus comprising means to implement any of Examples 1-20.

[0114] Example 23 is a system to implement any of Examples 1-20.

[0115] Example 24 is a method to implement any of Examples 1-20.

[0116] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the inventive subject matter can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.

[0117] In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” can include “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of

the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” and the like are used merely as labels, and are not intended to impose numerical requirements on their objects.

[0118] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) can be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features can be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter can lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the inventive subject matter should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

## Claims

1. A memory device comprising: a volatile memory; a memory array comprising: a plurality of prime rows; and a set of redundant rows; a memory controller configured to: receive a first memory access command comprising a first address for the memory array; determine that a first page of a first row for the first address comprises defective data; determine that a portion of the first row does not comprise defective data; access the first page from a page-based redundant row; receive a second memory access command comprising a second address for the memory array; determine that a second row for the second address entirely comprises defective data; and access a second page from a row-based redundant row.
2. The memory device of claim 1, wherein the determining that the first page of the first row for the first address comprises defective data is based on a lookup table that stores a section and row portion of addresses for which the corresponding data rows are at least partially defective.
3. The memory device of claim 1, wherein the determining that a portion of the first row does not comprise defective data is based on a soft post-package repair circuit that stores a complete address for the first page.
4. The memory device of claim 3, wherein the accessing of the first page from the page-based redundant row comprises accessing the first page from the complete address stored by the soft post-package repair circuit.
5. The memory device of claim 1, wherein the page-based redundant row stores replacement data for a third page of a third row.
6. The memory device of claim 5, wherein the third row is a row-based redundant row.
7. The memory device of claim 1, wherein a second page-based redundant row stores replacement data for a third page of the first row.
8. A method comprising: receiving, by a memory controller, a first memory access command comprising a first address for a memory array; determining that a first page of a first row for the first address comprises defective data; determining that a portion of the first row does not comprise defective data; accessing the first page from a page-based redundant row; receiving a second memory access command comprising a second address for the memory array; determining that a second row for the second address entirely comprises defective data; and accessing a second page

from a row-based redundant row.

- 9.** The method of claim 8, wherein the determining that the first page of the first row for the first address comprises defective data is based on a lookup table that stores a section and row portion of addresses for which the corresponding data rows are at least partially defective.
  - 10.** The method of claim 8, wherein the determining that a portion of the first row does not comprise defective data is based on a soft post-package repair circuit that stores a complete address for the first page.
  - 11.** The method of claim 10, wherein the accessing of the first page from the page-based redundant row comprises accessing the first page from the complete address stored by the soft post-package repair circuit.
  - 12.** The method of claim 8, wherein the page-based redundant row stores replacement data for a third page of a third row.
  - 13.** The method of claim 12, wherein the third row is a row-based redundant row.
  - 14.** The method of claim 8, wherein a second page-based redundant row stores replacement data for a third page of the first row.
  - 15.** A non-transitory machine-readable medium that stores instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising: receiving a first memory access command comprising a first address for a memory array; determining that a first page of a first row for the first address comprises defective data; determining that a portion of the first row does not comprise defective data; accessing the first page from a page-based redundant row; receiving a second memory access command comprising a second address for the memory array; determining that a second row for the second address entirely comprises defective data; and accessing a second page from a row-based redundant row.
  - 16.** The non-transitory machine-readable medium of claim 15, wherein the determining that the first page of the first row for the first address comprises defective data is based on a lookup table that stores a section and row portion of addresses for which the corresponding data rows are at least partially defective.
  - 17.** The non-transitory machine-readable medium of claim 15, wherein the determining that a portion of the first row does not comprise defective data is based on a soft post-package repair circuit that stores a complete address for the first page.
  - 18.** The non-transitory machine-readable medium of claim 17, wherein the accessing of the first page from the page-based redundant row comprises accessing the first page from the complete address stored by the soft post-package repair circuit.
  - 19.** The non-transitory machine-readable medium of claim 15, wherein the page-based redundant row stores replacement data for a third page of a third row.
  - 20.** The non-transitory machine-readable medium of claim 19, wherein the third row is a row-based redundant row.
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